

# HARIHARAN T S

AI / ML Engineer | Building ML-Integrated Applications

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[github.com/Tenyson-S](https://github.com/Tenyson-S) | [hariharan-ts-portfolio.vercel.app](https://hariharan-ts-portfolio.vercel.app)



## PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building machine learning-integrated applications using OCR, NLP, and backend APIs. Developed a real-time expense tracking system achieving 91% classification accuracy with sub-200ms inference latency. Familiar with FastAPI-based backend development and exploring LLM-powered features such as prompt-based insights and RAG pipelines. Focused on building practical, deployable AI solutions and continuously improving system performance.

## TECHNICAL SKILLS

**Programming Languages:** Python, Java

**AI / ML:** TensorFlow, Scikit-learn, CNNs, NLP, OCR (Tesseract), ML Inference.

**LLM Systems (Beginner Level):** RAG Pipelines (Learning Stage), Prompt Engineering, LLM Workflows

**Backend Development:** FastAPI (basic usage, API Development), REST API Design, Async I/O, Modular Pipeline Architecture, Request Validation, MySQL, SQLite, Firebase

**Frontend Development:** React.js, Component-based UI, Dynamic Data Rendering

**Tools & Platforms:** Git, GitHub, Postman, VS Code, Jupyter Notebook, Vercel

**Core CS:** Data Structures & Algorithms (400+ LeetCode), Object-Oriented Programming, System Design.

## PROJECTS

### AI-Powered Personal Finance Management System

2025 – Present

**Tech Stack:** Python · FastAPI · TensorFlow · Scikit-learn · Tesseract OCR · NLP · SQLite · Firebase · React · Prompt Engineering

- Engineered a full-stack AI application with a structured pipeline (SMS/receipt ingestion → OCR → NLP → ML classification → API) enabling real-time expense tracking with sub-200ms latency.
- Trained a multi-class ML classification model on 5,000+ transactions achieving ~91% accuracy using TensorFlow and Scikit-learn; implemented data augmentation to improve generalization.
- Designed a scalable backend with 12+ async FastAPI endpoints and decoupled ML microservices; implemented offline-first SQLite + Firebase sync architecture supporting 10,000+ records.
- Built a prompt-driven LLM insight layer for natural language budget summaries and financial recommendations using Prompt Engineering techniques.

### Interactive Data Explorer — One Piece Encyclopedia

2026 (Ongoing)

**Tech Stack:** React.js · JavaScript · Context API · Canvas API · JSON · Lazy Loading · Structured Data Modelling

- Designed a normalized data model representing 200+ characters and 30+ story arcs with relational mapping for efficient cross-entity access
- Implemented dynamic UI rendering using React with Context API to maintain state across 15+ routes
- Optimized performance by reducing initial load time by ~40% using lazy loading and memoization techniques
- Built interactive visualizations using Canvas API to display character relationships and story arc structures.

Traffic Sign Detection — CNN Classifier

2024

Tech Stack: Python · TensorFlow · CNN · OpenCV · Scikit-learn · GTSRB Dataset

- Trained a Convolutional Neural Network (CNN) on 50,000+ images across 43 classes achieving 96.4% test accuracy
- Applied data augmentation and class-weighted loss functions to handle class imbalance, reducing misclassification rate by 22%
- Evaluated model performance using precision, recall, and confusion matrix to ensure reliability for deployment scenarios

EDUCATION

B.E. Computer Science & Engineering (AI & ML) — Sethu Institute of Technology, Kariapatti 2022 – 2026

CGPA: 8.79 | Honors: Full Stack Development

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, Database Management Systems, Computer Vision

Class XII — Thiagarajar Model Higher Secondary School, Madurai 2022 | 84%

Class X — Thiagarajar Model Higher Secondary School, Madurai 2020 | 83.4%

CERTIFICATIONS

- AI Master Class (30-day Hands-on Program) — Novitech Pvt Ltd
- AI Primer Certification — Infosys Springboard
- Project-Based Robotics Training (2 programs) — Zetspire Technologies

ACHIEVEMENTS & AWARDS

- Department Topper — Awarded Academic Star & Above and Beyond honors for academic excellence.
- Winner in 7+ Inter-college Technical Symposiums — demonstrated technical depth across competitive events
- Top 5 — Hackathon at Mohammed Sathak Engineering College
- 400+ DSA problems solved on LeetCode across arrays, strings, trees, and dynamic programming

PERSONAL DETAILS

FULL NAME : HARIHARAN T S  
DATE OF BIRTH : 04/02/2005  
GENDER : MALE  
LANGUAGES KNOWN : TAMIL, SOURASHTRA, ENGLISH.  
NATIONALITY : INDIAN  
MARITAL STATUS : SINGLE